# UNIVERSITY LAB REPORT

Department of Electrical Engineering (Communications)

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| **Reg. No:** | 0672 | **Subject:** | computerPrograming |
| **Department:** | Electrical (Communications | **Date:** | May 2026 |

## Objectives

The objective of this lab is to practice fundamental string operations in C++. Students will learn how to use built-in string functions to calculate the length of a string and how to concatenate two strings to form a complete output. These are essential operations used in almost every real-world C++ application.

## Task 1: String Length Calculation

**Description:** In this task, a string variable named `message` is declared and initialized with a phrase. The built-in `.length()` function of the C++ string class is then used to find and display the total number of characters in that string.

### ■ C++ Source Code

```
// Header files ko include kar rahe hain
#include <iostream>
#include <string>
// std namespace use kar rahe hain taake cout aur cin likhne mein aasani ho
using namespace std;
int main() {
// String variable 'message' declare aur initialize kar rahe hain
string message = "Programming is the art of problem solving!";
// .length() function se string ki total characters count kar rahe hain
int len = message.length();
// Message string screen par print kar rahe hain
cout << "Message: " << message << endl;
// String ki length screen par display kar rahe hain
cout << "Length of the message: " << len << " characters" << endl;
```

```
return 0; // Program successfully khatam ho gaya
}
```

### Program Output

```
Message: Programming is the art of problem solving!
Length of the message: 42 characters
```

**Explanation:** The `string` class in C++ provides the `.length()` member function which returns the total count of characters stored in the string variable, including spaces and punctuation. Here the phrase contains 42 characters in total.

## Task 2: String Concatenation

**Description:** In this task, two separate string variables `firstName` and `lastName` are declared and initialized with a first and last name. These two strings are then joined (concatenated) using the + operator to form a complete full name, which is then displayed.

### C++ Source Code

```
// Zaruri header files include kar rahe hain
#include <iostream>
#include <string>
// Standard namespace declare kar rahe hain
using namespace std;
int main() {
// Pehla naam store karne ke liye string variable declare kar rahe hain
string firstName = "Shah";
// Akhri naam store karne ke liye doosra string variable declare kar rahe
hain
string lastName = "Khali";
// Dono strings ko ek saath jodne ke liye + operator use kar rahe hain
// Beech mein ek space bhi add kar rahe hain
string fullName = firstName + " " + lastName;
// Pehla naam alag print kar rahe hain
cout << "First Name : " << firstName << endl;
// Akhri naam alag print kar rahe hain
cout << "Last Name : " << lastName << endl;
// Pura naam (concatenated) screen par display kar rahe hain
cout << "Full Name : " << fullName << endl;
return 0; // Program successfully complete ho gaya
}
```

■ **Program Output**

```
First Name : Shah
Last Name : Khali
Full Name : Shah Khali
```

**Explanation:** The `+` operator in C++ is overloaded for the `string` class, which allows two strings to be concatenated directly. A space character `" "` is added between the first and last name to ensure proper formatting of the full name.

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## Conclusion

In this lab, two important string operations in C++ were successfully practiced. Task 1 demonstrated the use of the `.length()` function to calculate the number of characters in a given string. Task 2 showed how two separate string variables can be combined using the `+` operator to produce a meaningful output. Both programs were properly commented in Urdu (written in English letters) to explain each line of code clearly. These operations form the foundation of string manipulation, which is widely used in real-world software development.

Student Signature: ____________________ Instructor Signature: ____________________